

Project Notebook

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```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.tree import DecisionTreeClassifier
from sklearn.model_selection import train_test_split, GridSearchCV, KFold
from sklearn.metrics import roc_auc_score, accuracy_score, auc, roc_curve, ConfusionMatrixDisplay
from sklearn.linear_model import LogisticRegression
from utils import compute_accuracy, compute_auc

# Tests for compute_accuracy
assert compute_accuracy(np.array([1, 1, 0, 0]), np.array([1, 1, 0, 0])) == 1
assert compute_accuracy(np.array([1, 1, 0, 0]), np.array([1, 0, 1, 1])) == 1/4
assert compute_accuracy(np.array([1, 1, 0, 0]), np.array([1, 0, 0, 1])) == 1/2
assert compute_accuracy(np.array([1, 1, 0, 0]), np.array([1, 1, 0, 1])) == 3/4
print("Accuracy tests passed")

Accuracy tests passed

# Test for compute_auc
assert compute_auc(np.array([1, 1, 0, 0]), np.array([1, 0.9, 0.2, 0.1])) == 1
assert compute_auc(np.array([1, 0, 0, 0]), np.array([0.1, 0.02, 0.01, 0])) == 1
assert 0 <= compute_auc(np.array([1, 1, 0, 0]), np.array([1, 0.2, 0.5, 0.6])) <= 1
assert 0 <= compute_auc(np.array([1, 1, 0, 0]), np.array([0.1, 0.2, 0.5, 0.6])) <= 1
print("AUC tests passed")

AUC tests passed

pd.set_option('display.max_columns', None)

df = pd.read_csv("Data/WA_Fn -UseC_ -Telco -Customer -Churn.csv")
df['TotalCharges'] = pd.to_numeric(df['TotalCharges'], errors='coerce')
df = df.dropna(subset=["TotalCharges"])
df.info()
```

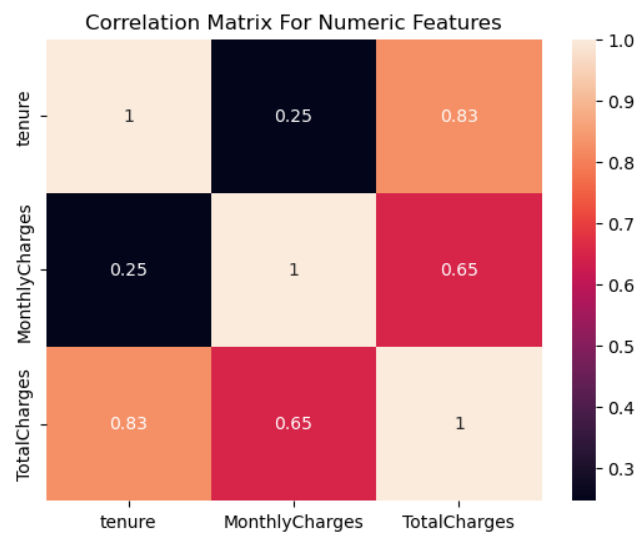
```

<class 'pandas.core.frame.DataFrame'>
Index: 7032 entries, 0 to 7042
Data columns (total 21 columns):
#   Column              Non -Null Count  Dtype
--  --
0   customerID          7032 non -null   object
1   gender              7032 non -null   object
2   SeniorCitizen       7032 non -null   int64
3   Partner             7032 non -null   object
4   Dependents          7032 non -null   object
5   tenure              7032 non -null   int64
6   PhoneService        7032 non -null   object
7   MultipleLines        7032 non -null   object
8   InternetService     7032 non -null   object
9   OnlineSecurity      7032 non -null   object
10  OnlineBackup        7032 non -null   object
11  DeviceProtection    7032 non -null   object
12  TechSupport         7032 non -null   object
13  StreamingTV         7032 non -null   object
14  StreamingMovies     7032 non -null   object
15  Contract            7032 non -null   object
16  PaperlessBilling    7032 non -null   object
17  PaymentMethod       7032 non -null   object
18  MonthlyCharges      7032 non -null   float64
19  TotalCharges        7032 non -null   float64
20  Churn               7032 non -null   object
dtypes: float64(2), int64(2), object(17)
memory usage: 1.2+ MB

numeric = df[['tenure', 'MonthlyCharges', 'TotalCharges']]
catagorical = df.drop(columns=['customerID', 'tenure', 'MonthlyCharges', 'TotalCharges'])

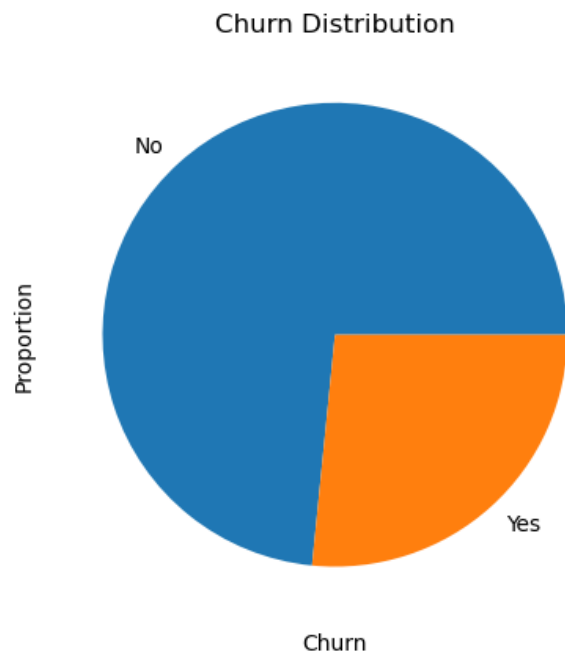
plt.title("Correlation Matrix For Numeric Features")
sns.heatmap(numeric.corr(), annot=True);

```



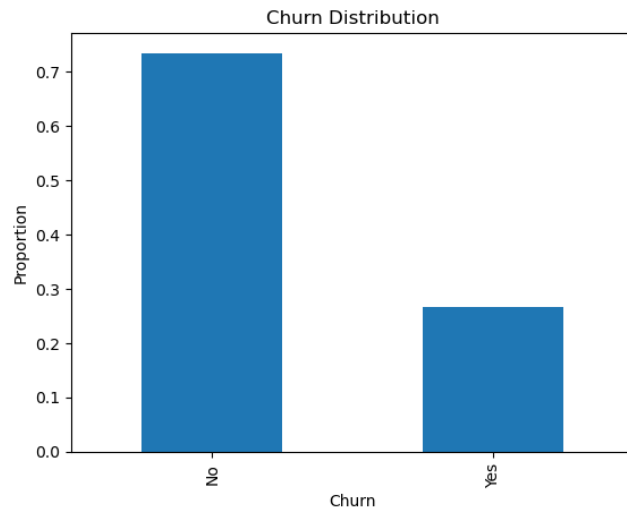
```
df["Churn"].value_counts(normalize=True).plot(kind='pie')
plt.title("Churn Distribution")
plt.xlabel("Churn")
plt.ylabel("Proportion");

plt.savefig("Churn_dist_pie.png", dpi=300, bbox_inches="tight")
```



```
df["Churn"].value_counts(normalize=True).plot(kind='bar')
plt.title("Churn Distribution")
plt.xlabel("Churn")
plt.ylabel("Proportion");

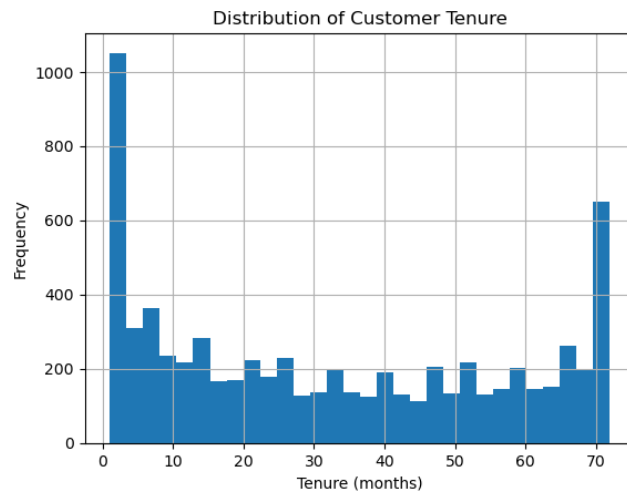
plt.savefig("Churn_dist_bar.png", dpi=300, bbox_inches="tight")
```



```
print(df["tenure"].describe())
df["tenure"].hist(bins=30)
plt.title("Distribution of Customer Tenure")
plt.xlabel("Tenure (months)")
plt.ylabel("Frequency");

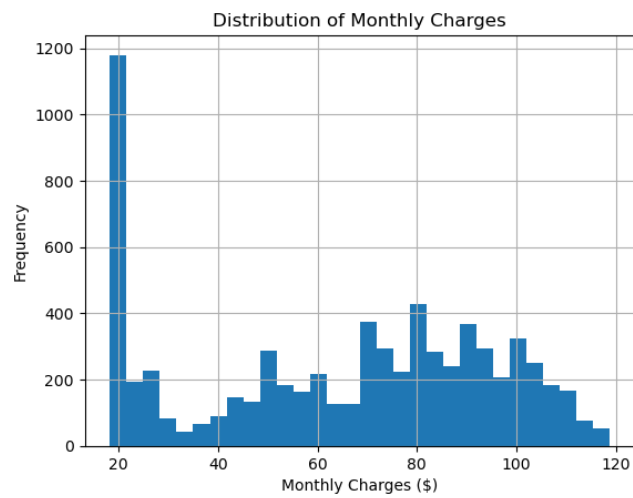
plt.savefig("Dist_customer_tenure.png", dpi=300, bbox_inches="tight")
```

```
count    7032.000000
mean      32.421786
std       24.545260
min        1.000000
25%        9.000000
50%       29.000000
75%       55.000000
max       72.000000
Name: tenure, dtype: float64
```



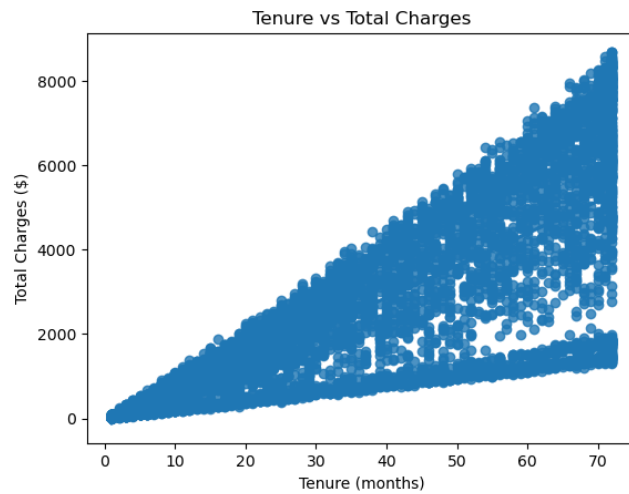
```
df["MonthlyCharges"].describe()
df["MonthlyCharges"].hist(bins=30)
plt.title("Distribution of Monthly Charges")
plt.xlabel("Monthly Charges ($)")
plt.ylabel("Frequency");

plt.savefig("Dist_monthly_charges.png", dpi=300, bbox_inches="tight")
```



```
sns.regplot(data=df, x="tenure", y="TotalCharges")
plt.title("Tenure vs Total Charges")
plt.xlabel("Tenure (months)")
plt.ylabel("Total Charges ($)");
```

```
plt.savefig("Tenure_vs_charges.png", dpi=300, bbox_inches="tight")
```

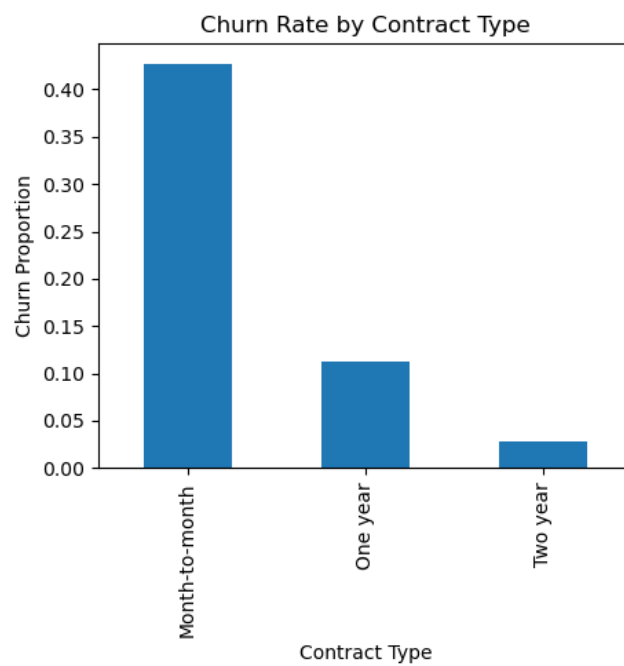
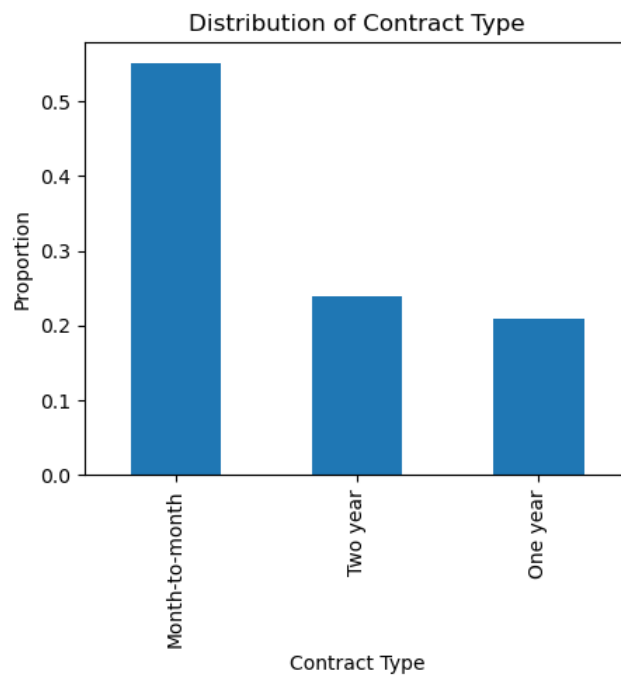


```
df["gender"].value_counts(normalize=True)
```

```
gender
Male      0.504693
Female    0.495307
Name: proportion, dtype: float64
```

```
plt.figure(figsize=(5,4))
catagorical["Contract"].value_counts(normalize=True).plot(kind="bar")
plt.title("Distribution of Contract Type")
plt.xlabel("Contract Type")
plt.ylabel("Proportion")
plt.savefig("Dist_contract_type.png", dpi=300, bbox_inches="tight")
plt.show()
```

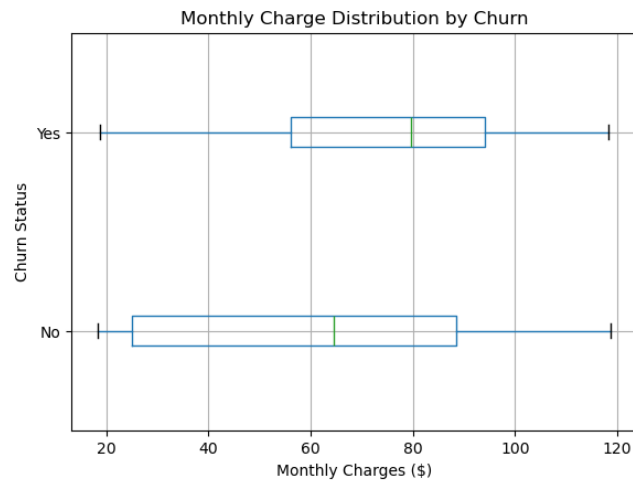
```
df["Churn_num"] = df["Churn"].map({"No":0, "Yes":1})
plt.figure(figsize=(5,4))
df.groupby("Contract")["Churn_num"].mean().plot(kind="bar")
plt.title("Churn Rate by Contract Type")
plt.xlabel("Contract Type")
plt.ylabel("Churn Proportion")
plt.savefig("Churn_by_contract.png", dpi=300, bbox_inches="tight")
plt.show()
```



```
plt.figure(figsize=(6,4))  
df.boxplot(column="MonthlyCharges", by="Churn", vert=False)
```

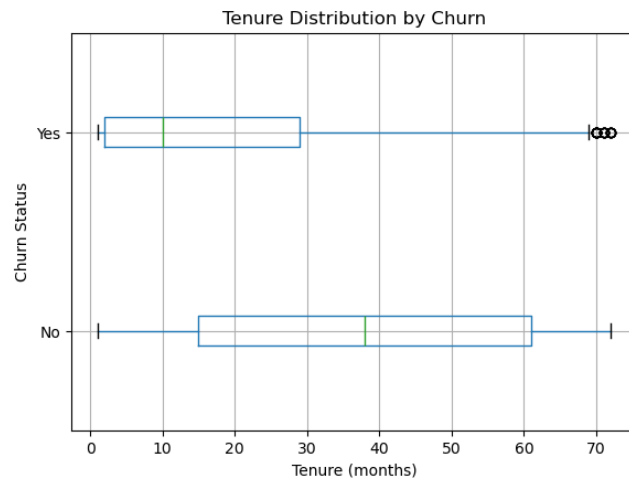
```
plt.title("Monthly Charge Distribution by Churn")
plt.suptitle("")
plt.xlabel("Monthly Charges ($)")
plt.ylabel("Churn Status")
plt.savefig("Monthly_charge_by_churn.png", dpi=300, bbox_inches="tight")
plt.show();
```

<Figure size 600x400 with 0 Axes>



```
plt.figure(figsize=(6,4))
df.boxplot(column="tenure", by="Churn", vert=False)
plt.title("Tenure Distribution by Churn")
plt.suptitle("")
plt.xlabel("Tenure (months)")
plt.ylabel("Churn Status")
plt.savefig("Tenure_by_churn.png", dpi=300, bbox_inches="tight")
plt.show()
```

<Figure size 600x400 with 0 Axes>



```
X = df.drop(columns=["customerID", "Churn", "Churn_num"])
X = pd.get_dummies(X, drop_first=True)
y = df["Churn_num"]
```

```
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.20, random_state=1, st
```

```
logistic_model = LogisticRegression(max_iter=2000).fit(X_train, y_train)
y_pred_logistic = logistic_model.predict(X_test)
logistic_model_acc = compute_accuracy(y_test, y_pred_logistic)
print(f'Logistic Regression Test Accuracy: {logistic_model_acc:.4f}')
```

Logistic Regression Test Accuracy: 0.7982

```
/srv/conda/envs/notebook/lib/python3.12/site -packages/sklearn/linear_model/_logistic.py:470
STOP: TOTAL NO. OF ITERATIONS REACHED LIMIT
```

Increase the number of iterations to improve the convergence (max_iter=2000).

You might also want to scale the data as shown in:

<https://scikit-learn.org/stable/modules/preprocessing.html>

Please also refer to the documentation for alternative solver options:

https://scikit-learn.org/stable/modules/linear_model.html#logistic-regression

```
n_iter_i = _check_optimize_result(
```

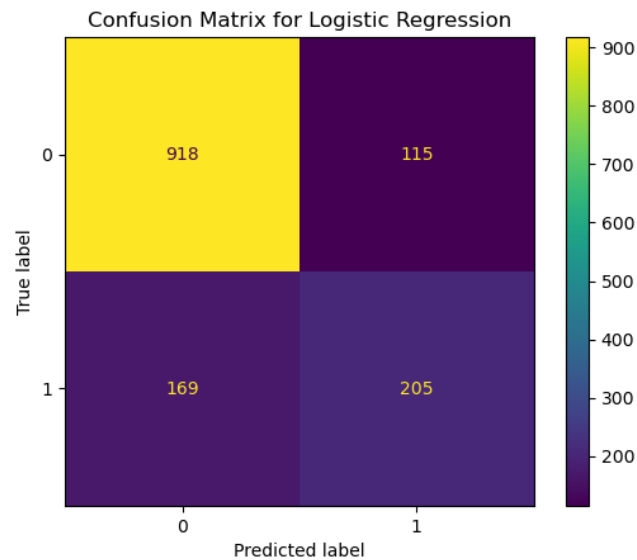
```
assert logistic_model.classes_[1] == 1
y_prob_logistic = logistic_model.predict_proba(X_test)[: , 1]
logistic_model_auc = compute_auc(y_test, y_prob_logistic)
print(f'Logistic Regression Test AUC: {logistic_model_auc:.4f}')
```

Logistic Regression Test AUC: 0.8308

```

ConfusionMatrixDisplay.from_predictions(y_test, y_pred_logistic)
plt.title("Confusion Matrix for Logistic Regression")
plt.savefig("Confusion_matrix_logistic_regression.png", dpi=300, bbox_inches="tight")
plt.show()

```



```

classification_tree = DecisionTreeClassifier(random_state=88)

grid_values = {'ccp_alpha': np.linspace(0, 0.035, 100)}
cv = KFold(n_splits=5, random_state=88, shuffle=True)

classification_tree_cv = GridSearchCV(classification_tree, param_grid=grid_values, \
                                       scoring='accuracy', cv=cv, verbose=0)
classification_tree_cv.fit(X_train, y_train)

ccp_alphas = classification_tree_cv.cv_results_['param_ccp_alpha'].data.astype(float)
accuracy_scores = classification_tree_cv.cv_results_['mean_test_score']

plt.figure(figsize=(8, 6))
plt.xlabel('ccp_alpha', fontsize=16)
plt.ylabel('CV Accuracy', fontsize=16)
plt.scatter(ccp_alphas, accuracy_scores, s=30)
plt.plot(ccp_alphas, accuracy_scores, linewidth=3)
plt.grid(True, which='both')
plt.tight_layout()
plt.savefig("Cross_validation_vs_ccp.png", dpi=300, bbox_inches="tight")
plt.show()

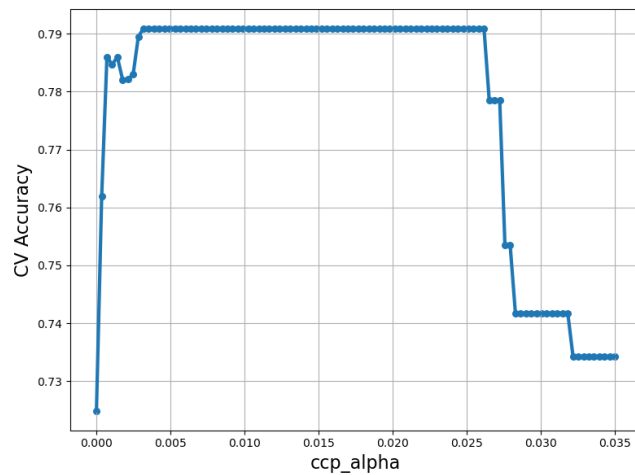
```

```

classification_tree = classification_tree_cv.best_estimator_
y_pred_tree = classification_tree.predict(X_test)

tree_acc = compute_accuracy(y_test, y_pred_tree)
tree_best_ccp_alpha = classification_tree_cv.best_params_['ccp_alpha']
print(f'CV CART Test Accuracy: {tree_acc:.4f}')
print(f'Best ccp_alpha: {tree_best_ccp_alpha:.4f}')

```



```

CV CART Test Accuracy: 0.7818
Best ccp_alpha: 0.0032

```

```

assert classification_tree.classes_[1] == 1
y_prob_tree = classification_tree.predict_proba(X_test)[: , 1]
tree_model_auc = compute_auc(y_test, y_prob_tree)
print(f'Classification Tree Test AUC: {tree_model_auc:.4f}')

```

```

Classification Tree Test AUC: 0.7937

```

```

ConfusionMatrixDisplay.from_predictions(y_test, y_pred_tree)
plt.title("Confusion Matrix for Classification Tree")
plt.savefig("Confusion_matrix_classification_tree.png", dpi=300, bbox_inches="tight")
plt.show()

```

